KCE PATHWAYS · GRADES 4–5 · UNIT 5: CHARTING THE COURSE · WEEK 21 · MATH

The Yellowstone Trail Map — Teacher Lesson Plan & Answer Key

Learning objectives: Students name and plot points on a coordinate plane with ordered pairs (x first, then y), move on the grid and find distances along grid lines with a map scale, read real-world graphs of ordered pairs, generate two patterns from two rules, form ordered pairs from corresponding terms, and describe the relationship between the patterns (for example, Rule B is always 2 times Rule A).

Duration: 35–45 minutes
Materials: Computer or tablet with the lesson open · pencil · printed Pre & Post sheet

Lesson flow (as it appears on screen)

1. Slide 0 — Intro and name: meet Quinlan (10), who is spending a week in Yellowstone National Park with his cousin Ramona (11). Each morning Ranger Roderick gives them a Junior Ranger map lesson on a grid laid over a trail map. The student types first and last name so results are sent. Read the scoring rules together: 2 tries per question, Pathway Points only for a correct first try, and the answer plus the reason appear after the 2nd miss. Typed number answers ignore capital letters, commas in large numbers, dollar signs and a final period.
2. Slide 1 — Warm-Up, Part 1: Volume Review (Unit 4) (4 pts, 2 each): V = l × w × h and V = B × h. If a student misses both warm-up slides, rework one volume on paper before moving on.
3. Slide 2 — Warm-Up, Part 2 (4 pts, 2 each): a missing height (volume ÷ base area) and a composite figure (add the two prisms, or work backward).
4. Slide 3 — Pencil Break (Pre): the student opens or prints the Pre/Post sheet (grade4-5/prepost/KCE_Math_PrePost_G4-5W21.pdf) and completes Page 1 (Pre-Lesson Warm-Up) off screen before any teaching.
5. Slide 4 — Why Rangers Love a Good Grid: nonfiction connection about Yellowstone, the world's first national park (March 1, 1872), with more than 10,000 hydrothermal features and more than 1,100 miles of trails. No questions.
6. Slide 5 — Picture It: The Coordinate Plane. Two number lines cross at the origin, (0, 0). In (2, 5) the x-coordinate says move 2 right and the y-coordinate says move 5 up. Order matters: (5, 2) is a different point. Memory trick: walk before you climb.
7. Slide 6 — Your Turn, Part 1: Naming Points (8 pts, 4 each): name a point on Ranger Roderick's practice grid (points A to G) and find the point at an ordered pair, including points on an axis.
8. Slide 7 — Your Turn, Part 2: Moving on the Grid (4 pts): start at a point, then move. Right and left change x; up and down change y.
9. Slide 8 — Teach: Real-World Graphs and Grid Distances. (hours hiked, miles walked): (3, 6) means 6 miles after 3 hours. Points that share a y-coordinate: subtract the x-coordinates (and the reverse), then multiply by the map scale.
10. Slide 9 — Your Turn, Part 1: Reading Real-World Graphs (8 pts, 4 each): miles walked between two hours, and the point that matches a sentence (x is hours, y is miles).
11. Slide 10 — Your Turn, Part 2: Grid Distance and Scale (4 pts): grid units times a 1/2-mile or 1/4-mile scale. Fractions, mixed numbers and decimals are accepted.

12. Slide 11 — Teach: Two Rules, Two Patterns. Quinlan earns 3 stamps a day and Ramona 6; corresponding terms make the pairs (0, 0), (3, 6), (6, 12), (9, 18), (12, 24). Every Ramona number is 2 times Quinlan's.
13. Slide 12 — Your Turn, Part 1: Two Rules (8 pts, 4 each): find a corresponding term, and choose the relationship that is true for EVERY pair ("times," not "more").
14. Slide 13 — Your Turn, Part 2: Making Ordered Pairs (4 pts): the next (Rule A, Rule B) pair.
15. Slide 14 — Vocabulary Round-Up: 8 flip cards (coordinate plane, x-axis, y-axis, origin, ordered pair, x-coordinate, y-coordinate, corresponding terms).
16. Slide 15 — Your Turn: Using the Words (6 pts, 3 each): one multiple-choice and one typed vocabulary item.
17. Slide 16 — Teach: Graphing Pattern Pairs. The stamp pairs make a straight line from the origin. Predict (30 × 2 = 60), work backward (42 ÷ 2 = 21), and notice that some relationships add instead of multiply (always 3 more).
18. Slide 17 — Your Turn: Thinking With Patterns (6 pts, 3 each): predict a far-off corresponding term, and choose the matching list of pairs or the true relationship.
19. Slide 18 — Real World: Ranger Roderick's Map (4 pts): a scaled grid walk, a bison count, or the stamp table worked backward.
20. Slide 19 — Real World: Trail Data (4 pts): water bottles, a there-and-back walk, or photo patterns.
21. Slide 20 — Think It Through: Explain and Find the Mistake (10 pts, 5 each): a "which explanation is correct?" item and a "find the mistake" item. Ask the student to say why each wrong choice is wrong.
22. Slide 21 — Fix the Sentence, Part 1 (4 pts, 2 each): typed sentences checked exactly (capital letters, quotation marks, commas, end marks, and one grammar fix). Straight or curly quotes and extra spaces are OK. The corrected sentence appears after the 2nd miss.
23. Slide 22 — Fix the Sentence, Part 2 (4 pts, 2 each): two more sentences, same rules. Both serial-comma and no-serial-comma forms of a list are accepted where a list appears.
24. Slide 23 — Match-Up Round (10 pts): 5 grid moves or pattern rules matched to ordered pairs or relationships, 2 pts each on the first try. Watch the order ("6 right, 2 up" is (6, 2), not (2, 6)).
25. Slide 24 — Speed Trail (bonus XP only): 3 quick items against the student's own best time.
26. Slide 25 — Looking Back reflection (bonus XP only).
27. Slide 26 — Boss Battle: The Junior Ranger Map Challenge (8 pts, 4 each): a loop hike, two hiking groups as pattern pairs, or a filling water tank.
28. Slide 27 — Pencil Break (Post): the student completes Page 2 (Post-Lesson Challenge) of the same sheet (grade4-5/prepost/KCE_Math_PrePost_G4-5W21.pdf), plus the ★ Challenge if time allows, before seeing the score.
29. Slide 28 — Results (Map Lesson Complete!): the score out of 100 is sent to KingCretot Experience. 85 or higher shows "Yellowstone Map Master!" with the congratulations message "You charted the whole practice map in Yellowstone!" and a big confetti burst; below 85 shows "Good Work on the Trail!" with encouragement ("You are getting closer every time."), and the next attempt brings a fresh question set (sets rotate 1, 2, 3). Both messages remind the student to finish the Post-Lesson Challenge sheet.

Lesson answer key with reasoning

Every scored item the student sees, in order, across all three rotating question sets. Each item allows 2 tries and earns Pathway Points only on a correct first try. The "why" is the same reasoning shown on screen after a second missed try.

WARM-UP: VOLUME REVIEW (UNIT 4) (2 POINTS EACH)

ID	Question	Answer	Why (explain to student)
r1_s0	Set 1: Volume (Week 19): a bear-proof food storage box at a campsite is 4 feet long, 3 feet wide, and 3 feet tall. What is its volume in cubic feet? Cubic feet:	**36** (also accepted with a unit word, such as 36 cubic feet or 36 cubic units, or 6 other unit spellings)	V = l × w × h = 4 × 3 × 3. 4 × 3 = 12, and 12 × 3 = 36 cubic feet.
r2_s0	Set 1: Volume (Week 19): the base of a cooler has an area of 54 square inches. The cooler is 11 inches tall. What is its volume? [594 cubic inches / 65 cubic inches / 540 cubic inches / 5,940 cubic inches]	**594 cubic inches**	V = B × h = 54 × 11 = 594 cubic inches. (65 comes from adding 54 + 11.)
r3_s0	Set 1: Two steps (Week 19): a storage shed holds 315 cubic feet. It is 9 feet long and 5 feet wide. How tall is it? Height (feet):	**7** (also accepted: 7 feet, 7 foot, 7 ft)	Area of the base: 9 × 5 = 45 square feet. Height: 315 ÷ 45 = 7 feet. Check: 9 × 5 × 7 = 315.
r4_s0	Set 1: Composite figure (Week 19): a stone bench is made of two prisms. Prism A is 6 feet by 3 feet by 2 feet. Prism B is 2 feet by 3 feet by 4 feet. What is the total volume? Cubic feet:	**60** (also accepted with a unit word, such as 60 cubic feet or 60 cubic units, or 6 other unit spellings)	Prism A: 6 × 3 × 2 = 36. Prism B: 2 × 3 × 4 = 24. Add: 36 + 24 = 60 cubic feet.
r1_s1	Set 2: Volume (Week 19): a supply chest at the ranger station is 7 feet long, 4 feet wide, and 3 feet tall. What is its volume in cubic feet? Cubic feet:	**84** (also accepted with a unit word, such as 84 cubic feet or 84 cubic units, or 6 other unit spellings)	V = l × w × h = 7 × 4 × 3. 7 × 4 = 28, and 28 × 3 = 84 cubic feet.
r2_s1	Set 2: Volume (Week 19): the base of a backpack cooler has an area of 46 square inches. It is 12 inches tall. What is its volume? [552 cubic inches / 58 cubic inches / 460 cubic inches / 5,520 cubic inches]	**552 cubic inches**	V = B × h = 46 × 12 = 552 cubic inches. (58 comes from adding 46 + 12.)
r3_s1	Set 2: Two steps (Week 19): a box of trail maps holds 432 cubic inches. It is 12 inches long and 6 inches wide. How tall is it? Height (inches):	**6** (also accepted: 6 inches, 6 inch, 6 in)	Area of the base: 12 × 6 = 72 square inches. Height: 432 ÷ 72 = 6 inches. Check: 12 × 6 × 6 = 432.
r4_s1	Set 2: Composite figure (Week 19): a ranger station porch step is made of two prisms. Prism A is 8 feet by 4 feet by 2 feet. Prism B is 3 feet by 4 feet by 3 feet. What is the total volume? Cubic feet:	**100** (also accepted with a unit word, such as 100 cubic feet or 100 cubic units, or 6 other unit spellings)	Prism A: 8 × 4 × 2 = 64. Prism B: 3 × 4 × 3 = 36. Add: 64 + 36 = 100 cubic feet.
r1_s2	Set 3: Volume (Week 19): a firewood bin at a campground is 11 feet long, 3 feet wide, and 4 feet tall. What is its volume in cubic feet? Cubic feet:	**132** (also accepted with a unit word, such as 132 cubic feet or 132 cubic units, or 6 other unit spellings)	V = l × w × h = 11 × 3 × 4. 11 × 3 = 33, and 33 × 4 = 132 cubic feet.
r2_s2	Set 3: Volume (Week 19): the base of a map case has an area of 38 square inches. The case is 15 inches tall. What is its volume? [570 cubic inches / 53 cubic inches / 380 cubic inches / 5,700 cubic inches]	**570 cubic inches**	V = B × h = 38 × 15 = 570 cubic inches. (53 comes from adding 38 + 15.)
r3_s2	Set 3: Two steps (Week 19): a gear closet holds 378 cubic feet. It is 9 feet long and 6 feet wide. How tall is it? Height (feet):	**7** (also accepted: 7 feet, 7 foot, 7 ft)	Area of the base: 9 × 6 = 54 square feet. Height: 378 ÷ 54 = 7 feet. Check: 9 × 6 × 7 = 378.

ID	Question	Answer	Why (explain to student)
r4_s2	Set 3: Working backward (Week 19): a composite cabin has a total volume of 275 cubic feet. Prism A is 6 feet long, 5 feet wide, and 5 feet tall. What is the volume of Prism B? Cubic feet:	**125** (also accepted with a unit word, such as 125 cubic feet or 125 cubic units, or 6 other unit spellings)	Prism A: 6 × 5 × 5 = 150 cubic feet. Prism B: 275 − 150 = 125 cubic feet.

YOUR TURN: NAMING POINTS (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
l1_s0	Set 1: Type the ordered pair that names point A. Use the form (x, y). Point A:	**(3, 6)** (also accepted: 3, 6, x = 3, y = 6, x=3,y=6)	Point A is 3 units right of the origin and 6 units up, so it is (3, 6). The x-coordinate always comes first.
l2_s0	Set 1: Which point is located at (0, 4)? [Point C / Point D / Point A / Point B]	**Point C**	Start at the origin, move 0 units right, then 4 units up. That lands on point C, which sits on the y-axis.
l1_s1	Set 2: Type the ordered pair that names point G. Use the form (x, y). Point G:	**(8, 5)** (also accepted: 8, 5, x = 8, y = 5, x=8,y=5)	Point G is 8 units right of the origin and 5 units up, so it is (8, 5). The x-coordinate always comes first.
l2_s1	Set 2: Which point is located at (5, 0)? [Point D / Point C / Point G / Point B]	**Point D**	Start at the origin, move 5 units right, then 0 units up. That lands on point D, which sits on the x-axis.
l1_s2	Set 3: Type the ordered pair that names point F. Use the form (x, y). Point F:	**(2, 1)** (also accepted: 2, 1, x = 2, y = 1, x=2,y=1)	Point F is 2 units right of the origin and 1 unit up, so it is (2, 1). The x-coordinate always comes first.
l2_s2	Set 3: Which point lies ON the x-axis? [Point D / Point C / Point F / Point B]	**Point D**	A point on the x-axis has a y-coordinate of 0. Point D is (5, 0). Point C is (0, 4), which is on the y-axis.

YOUR TURN: MOVING ON THE GRID (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
l3_s0	Set 1: Two steps: Ramona starts at the trailhead at (2, 3). She walks 4 units right and then 3 units up. What ordered pair names the place where she stops? Ordered pair:	**(6, 6)** (also accepted: 6, 6, x = 6, y = 6, x=6,y=6)	Start at (2, 3). Moving right adds to x: 2 + 4 = 6. Moving up adds to y: 3 + 3 = 6. She stops at (6, 6).
l3_s1	Set 2: Two steps: Quinlan starts at the footbridge at (7, 2). He walks 5 units left and then 6 units up. What ordered pair names the place where he stops? Ordered pair:	**(2, 8)** (also accepted: 2, 8, x = 2, y = 8, x=2,y=8)	Start at (7, 2). Moving left subtracts from x: 7 − 5 = 2. Moving up adds to y: 2 + 6 = 8. He stops at (2, 8).
l3_s2	Set 3: Two steps: Ramona starts at the picnic area at (1, 6). She walks 6 units right and then 4 units down. What ordered pair names the place where she stops? Ordered pair:	**(7, 2)** (also accepted: 7, 2, x = 7, y = 2, x=7,y=2)	Start at (1, 6). Moving right adds to x: 1 + 6 = 7. Moving down subtracts from y: 6 − 4 = 2. She stops at (7, 2).

YOUR TURN: READING REAL-WORLD GRAPHS (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
k1_s0	Set 1: Two steps: Quinlan's group graphs (hours hiked, miles walked). The points are (1, 3), (2, 6), (3, 9), and (4, 12). How many miles did the group walk between hour 2 and hour 4? Miles:	**6** (also accepted: 6 miles, 6 mile, 6 mi)	At hour 2 the group had walked 6 miles, and at hour 4 they had walked 12 miles. 12 − 6 = 6 miles.
k2_s0	Set 1: On the same kind of graph, which point shows "after 5 hours, the group had walked 15 miles"? [(5, 15) / (15, 5) / (5, 10) / (0, 15)]	**(5, 15)**	The x-coordinate is the hours (5) and the y-coordinate is the miles (15), so the point is (5, 15). (15, 5) would mean 15 hours and only 5 miles.
k1_s1	Set 2: Two steps: Ramona's group graphs (hours hiked, miles walked). The points are (1, 4), (2, 8), (3, 12), and (5, 20). How many miles did the group walk from hour 1 to hour 5? Miles:	**16** (also accepted: 16 miles, 16 mile, 16 mi)	At hour 1 the group had walked 4 miles, and at hour 5 they had walked 20 miles. 20 − 4 = 16 miles.
k2_s1	Set 2: On the same kind of graph, which point shows "after 6 hours, the group had walked 9 miles"? [(6, 9) / (9, 6) / (6, 15) / (3, 9)]	**(6, 9)**	The x-coordinate is the hours (6) and the y-coordinate is the miles (9), so the point is (6, 9). (9, 6) would mean 9 hours and 6 miles.
k1_s2	Set 3: Two steps: Ranger Roderick's group graphs (hours hiked, miles walked). The points are (0, 0), (2, 5), (4, 10), and (6, 15). How many miles did the group walk between hour 2 and hour 6? Miles:	**10** (also accepted: 10 miles, 10 mile, 10 mi)	At hour 2 the group had walked 5 miles, and at hour 6 they had walked 15 miles. 15 − 5 = 10 miles.
k2_s2	Set 3: On the same kind of graph, which point shows "after 4 hours, the group had walked 7 miles"? [(4, 7) / (7, 4) / (4, 11) / (0, 7)]	**(4, 7)**	The x-coordinate is the hours (4) and the y-coordinate is the miles (7), so the point is (4, 7). (7, 4) would mean 7 hours and 4 miles.

YOUR TURN: GRID DISTANCE AND SCALE (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
k3_s0	Set 1: Two steps: on the practice map, the Ranger Station is at (2, 6) and the Footbridge is at (9, 6). Each grid unit stands for 1/2 mile. How many miles apart are they along the straight grid line? Miles:	**3 1/2** (also accepted, with or without the unit word: 7/2, 3 and 1/2, 3.5)	Both points have a y-coordinate of 6, so subtract the x-coordinates: 9 − 2 = 7 units. Each unit is 1/2 mile: 7 × 1/2 = 7/2 = 3 1/2 miles.
k3_s1	Set 2: Two steps: on the practice map, the campsite is at (4, 1) and the overlook is at (4, 9). Each grid unit stands for 1/4 mile. How many miles apart are they along the straight grid line? Miles:	**2** (also accepted: 2 miles, 2 mile, 2 mi)	Both points have an x-coordinate of 4, so subtract the y-coordinates: 9 − 1 = 8 units. Each unit is 1/4 mile: 8 × 1/4 = 2 miles.
k3_s2	Set 3: Three steps: the picnic area is at (1, 3) and the boardwalk is at (7, 3). Each grid unit stands for 1/2 mile. Ramona walks from the picnic area to the boardwalk and back the same way. How many miles does she walk in all? Miles:	**6** (also accepted: 6 miles, 6 mile, 6 mi)	Both points have a y-coordinate of 3, so subtract: 7 − 1 = 6 units. 6 × 1/2 = 3 miles one way. There and back is 3 × 2 = 6 miles.

YOUR TURN: TWO RULES (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
e1_s0	Set 1: Rule A: start at 0, add 4. Rule B: start at 0, add 12. Write the first 5 numbers of each pattern. Which Rule B number corresponds to 16 in Rule A? Rule B number:	**48**	Rule A: 0, 4, 8, 12, 16. Rule B: 0, 12, 24, 36, 48. 16 is the 5th number of Rule A, and the 5th number of Rule B is 48. (Rule B is always 3 times Rule A: 16 × 3 = 48.)
e2_s0	Set 1: Rule A: start at 0, add 5. Rule B: start at 0, add 10. Which statement is true for EVERY pair of corresponding terms? [Each Rule B number is 2 times the Rule A number / Each Rule B number is 5 more than the Rule A number / Each Rule A number is 2 times the Rule B number / Each Rule B number is 10 more than the Rule A number]	**Each Rule B number is 2 times the Rule A number**	Rule A: 0, 5, 10, 15. Rule B: 0, 10, 20, 30. Each Rule B number is 2 times the Rule A number (10 = 2 × 5). "5 more" works for (5, 10) but not for (10, 20).
e1_s1	Set 2: Rule A: start at 0, add 2. Rule B: start at 0, add 10. Write the first 5 numbers of each pattern. Which Rule B number corresponds to 8 in Rule A? Rule B number:	**40**	Rule A: 0, 2, 4, 6, 8. Rule B: 0, 10, 20, 30, 40. 8 is the 5th number of Rule A, and the 5th number of Rule B is 40. (Rule B is always 5 times Rule A.)
e2_s1	Set 2: Rule A: start at 0, add 6. Rule B: start at 0, add 3. Which statement is true for EVERY pair of corresponding terms? [Each Rule B number is half of the Rule A number / Each Rule B number is 3 less than the Rule A number / Each Rule B number is 2 times the Rule A number / Each Rule B number equals the Rule A number]	**Each Rule B number is half of the Rule A number**	Rule A: 0, 6, 12, 18. Rule B: 0, 3, 6, 9. Each Rule B number is half of the Rule A number. "3 less" works for (6, 3) but not for (12, 6).
e1_s2	Set 3: Rule A: start at 0, add 5. Rule B: start at 0, add 15. Write the first 5 numbers of each pattern. Which Rule B number corresponds to 20 in Rule A? Rule B number:	**60**	Rule A: 0, 5, 10, 15, 20. Rule B: 0, 15, 30, 45, 60. 20 is the 5th number of Rule A, and the 5th number of Rule B is 60. (Rule B is always 3 times Rule A.)
e2_s2	Set 3: Rule A: start at 0, add 2. Rule B: start at 0, add 8. Which statement is true for EVERY pair of corresponding terms? [Each Rule B number is 4 times the Rule A number / Each Rule B number is 6 more than the Rule A number / Each Rule B number is 2 times the Rule A number / Each Rule B number is 8 times the Rule A number]	**Each Rule B number is 4 times the Rule A number**	Rule A: 0, 2, 4, 6. Rule B: 0, 8, 16, 24. Each Rule B number is 4 times the Rule A number, because 8 = 4 × 2. "6 more" works for (2, 8) but not for (4, 16).

YOUR TURN: MAKING ORDERED PAIRS (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
e3_s0	Set 1: Rule A: start at 0, add 3. Rule B: start at 0, add 9. The ordered pairs are (Rule A, Rule B). Which ordered pair comes right after (9, 27)? Ordered pair:	**(12, 36)** (also accepted: 12, 36, x = 12, y = 36, x=12,y=36)	Rule A goes 0, 3, 6, 9, 12 and Rule B goes 0, 9, 18, 27, 36. The pair after (9, 27) is (12, 36).
e3_s1	Set 2: Rule A: start at 0, add 4. Rule B: start at 0, add 8. The ordered pairs are (Rule A, Rule B). Which ordered pair comes right after (12, 24)? Ordered pair:	**(16, 32)** (also accepted: 16, 32, x = 16, y = 32, x=16,y=32)	Rule A goes 0, 4, 8, 12, 16 and Rule B goes 0, 8, 16, 24, 32. The pair after (12, 24) is (16, 32).

ID	Question	Answer	Why (explain to student)
e3_s2	Set 3: Rule A: start at 0, add 6. Rule B: start at 0, add 18. The ordered pairs are (Rule A, Rule B). Which ordered pair comes right after (18, 54)? Ordered pair:	**(24, 72)** (also accepted: 24, 72, x = 24, y = 72, x=24,y=72)	Rule A goes 0, 6, 12, 18, 24 and Rule B goes 0, 18, 36, 54, 72. The pair after (18, 54) is (24, 72).

YOUR TURN: USING THE WORDS (3 POINTS EACH)

ID	Question	Answer	Why (explain to student)
v1_s0	Set 1: From the lesson: “Every trip on the grid starts at the origin.” Which ordered pair names the origin? [(0, 0) / (1, 1) / (0, 1) / (1, 0)]	**(0, 0)**	The origin is where the x-axis and y-axis meet. You have not moved right or up, so it is (0, 0).
v2_s0	Set 1: Type your answer: the first number in an ordered pair is called the ___-coordinate. (one letter)	**x** (also accepted: x-coordinate, x coordinate, the x-coordinate)	The first number in an ordered pair is the x-coordinate. It tells how far to move right.
v1_s1	Set 2: From the lesson: “The flat one is the ___.” Which word names the horizontal number line on a coordinate plane? [x-axis / y-axis / origin / ordered pair]	**x-axis**	The x-axis is the horizontal (flat) number line. The y-axis goes straight up.
v2_s1	Set 2: Type your answer: “An ordered ___ such as (2, 5) tells you how to get to a point.” (one word)	**pair** (also accepted: ordered pair; with “a” or “an” or in the plural)	Two numbers written in a set order, like (2, 5), make an ordered pair. Switching the order names a different point.
v1_s2	Set 3: From the lesson: “The 3rd number of each pattern is 6 and 12.” Numbers in the same position in two patterns are called ___. [Corresponding terms / Ordered pairs / Coordinates / Axes]	**Corresponding terms**	Corresponding terms sit in the same position in two patterns, like the 3rd number of each. Together they make one ordered pair.
v2_s2	Set 3: Type your answer: the number line that goes straight up on a coordinate plane is called the ___-axis. (one letter)	**y** (also accepted: y-axis, y axis, the y-axis)	The number line that goes straight up (vertical) is the y-axis.

YOUR TURN: THINKING WITH PATTERNS (3 POINTS EACH)

ID	Question	Answer	Why (explain to student)
t1_s0	Set 1: Rule A: start at 0, add 7. Rule B: start at 0, add 14. Without listing every number, which Rule B number corresponds to 70 in Rule A? Rule B number:	**140**	Rule B adds 14 and Rule A adds 7, so each Rule B number is 2 times the Rule A number. 70 × 2 = 140.
t2_s0	Set 1: Which list of ordered pairs comes from Rule A: start at 0, add 2, and Rule B: start at 0, add 5? [(0, 0), (2, 5), (4, 10), (6, 15) / (0, 0), (5, 2), (10, 4), (15, 6) / (0, 0), (2, 5), (4, 8), (6, 11) / (2, 5), (3, 6), (4, 7), (5, 8)]	**(0, 0), (2, 5), (4, 10), (6, 15)**	Rule A gives 0, 2, 4, 6 and Rule B gives 0, 5, 10, 15. Pair them in order (Rule A, Rule B): (0, 0), (2, 5), (4, 10), (6, 15). The second choice switches the order.
t1_s1	Set 2: Rule A: start at 0, add 3. Rule B: start at 0, add 12. Without listing every number, which Rule B number corresponds to 45 in Rule A? Rule B number:	**180**	Rule B adds 12 and Rule A adds 3, so each Rule B number is 4 times the Rule A number. 45 × 4 = 180.

ID	Question	Answer	Why (explain to student)
t2_s1	Set 2: Rule A: start at 2, add 2. Rule B: start at 5, add 2. Which statement is true for EVERY pair of corresponding terms? [Each Rule B number is 3 more than the Rule A number / Each Rule B number is 2 times the Rule A number / Each Rule B number is 5 more than the Rule A number / Each Rule B number is half of the Rule A number]	**Each Rule B number is 3 more than the Rule A number**	Rule A: 2, 4, 6, 8. Rule B: 5, 7, 9, 11. Both add 2, so the gap never changes: each Rule B number is 3 more. "2 times" fails, because 2 × 2 = 4, not 5.
t1_s2	Set 3: Rule A: start at 0, add 8. Rule B: start at 0, add 2. Without listing every number, which Rule B number corresponds to 48 in Rule A? Rule B number:	**12**	Rule B adds 2 and Rule A adds 8, so each Rule B number is 1/4 of the Rule A number. 48 ÷ 4 = 12.
t2_s2	Set 3: The points (1, 4), (2, 8), and (3, 12) are on a graph made from two patterns. If the pattern continues, which point is also on the graph? [(5, 20) / (5, 16) / (20, 5) / (4, 15)]	**(5, 20)**	Each y-coordinate is 4 times the x-coordinate: 1 × 4 = 4, 2 × 4 = 8, 3 × 4 = 12. So x = 5 goes with y = 5 × 4 = 20.

REAL WORLD: RANGER RODERICK'S MAP (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
w1_s0	Set 1: *The Situation Ranger Roderick's practice map shows the Ranger Station at (1, 2), the Picnic Area at (1, 8), and the Overlook at (6, 8). Each grid unit stands for 1/4 mile. Quinlan walks along the grid lines from the Ranger Station to the Picnic Area, and then on to the Overlook.* Three steps: how many miles does Quinlan walk in all? Miles:	**2 3/4** (also accepted, with or without the unit word: 11/4, 2 and 3/4, 2.75)	Ranger Station to Picnic Area: 8 − 2 = 6 units. Picnic Area to Overlook: 6 − 1 = 5 units. Total: 6 + 5 = 11 units. 11 × 1/4 = 11/4 = 2 3/4 miles.
w1_s1	Set 2: *The Situation Ramona counts the bison that cross a meadow, watching from a safe distance, well over 25 yards away. She records (minutes watched, total bison counted so far): (10, 6), (20, 12), and (30, 18). The pattern keeps going the same way.* Two steps: if the pattern continues, how many bison will Ramona have counted after 60 minutes? [36 bison / 24 bison / 60 bison / 42 bison]	**36 bison**	The count goes up 6 bison for every 10 minutes: (40, 24), (50, 30), (60, 36). Or think of it this way: 60 minutes is 6 groups of 10 minutes, and 6 × 6 = 36 bison.
w1_s2	Set 3: *The Situation Quinlan and Ramona both start with 0 Junior Ranger stamps. Quinlan earns 3 stamps each day and Ramona earns 6 stamps each day, just like the table in the lesson.* Two steps: on the day Ramona has 42 stamps, how many stamps does Quinlan have? Stamps:	**21** (also accepted: 21 stamps, 21 stamp)	Ramona always has 2 times as many stamps as Quinlan. Working backward: 42 ÷ 2 = 21 stamps. (Check: Ramona reaches 42 on day 7, and Quinlan has 7 × 3 = 21.)

REAL WORLD: TRAIL DATA (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
w2_s0	Set 1: *The Situation At the visitor center, Ramona fills water bottles before a hike. She records (bottles filled, ounces of water): (1, 24), (2, 48), and (3, 72). Every bottle holds the same amount.* Two steps: which ordered pair shows the ounces of water in 7 bottles? [(7, 168) / (168, 7) / (7, 96) / (7, 144)]	**(7, 168)**	Each bottle holds 24 ounces. 7 × 24 = 168 ounces, and bottles come first, so the pair is (7, 168). (7, 96) adds 24 to 72 only once.
w2_s1	Set 2: *The Situation The practice map shows the Visitor Center at (3, 1), the Boardwalk at (3, 6), and the Footbridge at (9, 6). Each grid unit stands for 1/4 mile. Ramona walks along the grid lines from the Visitor Center to the Boardwalk to the Footbridge, and then back the same way.* Three steps: how many miles does Ramona walk in all? [5 1/2 miles / 2 3/4 miles / 22 miles / 11 miles]	**5 1/2 miles**	Visitor Center to Boardwalk: 6 − 1 = 5 units. Boardwalk to Footbridge: 9 − 3 = 6 units. One way: 11 units × 1/4 = 11/4 = 2 3/4 miles. There and back: 2 3/4 × 2 = 5 1/2 miles. (2 3/4 is only one way, and 22 forgets the scale.)
w2_s2	Set 3: *The Situation Quinlan makes two patterns about his trail photos. Rule A: start at 0, add 5. Rule B: start at 0, add 20. He graphs the ordered pairs (Rule A, Rule B).* Two steps: on Quinlan's graph, what is the y-coordinate of the point whose x-coordinate is 35? y-coordinate:	**140**	Rule B adds 20 and Rule A adds 5, so each y-coordinate is 4 times the x-coordinate. 35 × 4 = 140.

THINK IT THROUGH: EXPLAIN AND FIND THE MISTAKE (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
x1_s0	Set 1: Which explanation is correct? How do you plot the point (5, 2)? [Start at (0, 0), move 5 units right along the x-axis, then 2 units up / Start at (0, 0), move 5 units up, then 2 units right / Start at (0, 0), move 2 units right, then 5 units up / Start at (5, 5), then move 2 units down]	**Start at (0, 0), move 5 units right along the x-axis, then 2 units up**	The x-coordinate (5) comes first and tells how far right. The y-coordinate (2) tells how far up. Walk before you climb.
x2_s0	Set 1: Find the mistake: Rasmus plots the Footbridge at (6, 3) by moving up 6 units and then right 3 units. What did he do wrong? [He switched the order. He should move 6 units right, then 3 units up / He should move 3 units right and 6 units up / He should start at (6, 6) / Nothing. Up first is correct]	**He switched the order. He should move 6 units right, then 3 units up**	In (6, 3), the first number is the x-coordinate, so move 6 units right first. Then the y-coordinate tells you to move 3 units up. Rasmus actually plotted (3, 6).
x1_s1	Set 2: Which explanation is correct? Rule A: start at 0, add 3. Rule B: start at 0, add 6. Why do the graphed pairs make a straight line from the origin? [Each new pair is 3 units right and 6 units up from the one before / Both rules add 3 each time / The y-axis is longer than the x-axis / Because 3 + 6 = 9]	**Each new pair is 3 units right and 6 units up from the one before**	The pairs are (0, 0), (3, 6), (6, 12), (9, 18). Every step moves the same amount, 3 right and 6 up, so the points line up in a straight line.

ID	Question	Answer	Why (explain to student)
x2_s1	Set 2: Find the mistake: Ruthanne says the distance from (2, 4) to (2, 9) is 2 units, because both x-coordinates are 2. What should she fix? [The points are on a vertical line, so subtract the y-coordinates: 9 − 4 = 5 units / Add the y-coordinates: 4 + 9 = 13 units / Add all four numbers: 2 + 4 + 2 + 9 = 17 units / Nothing. 2 units is correct]	**The points are on a vertical line, so subtract the y-coordinates: 9 − 4 = 5 units**	When two points share the same x-coordinate, they sit on the same vertical line. The distance is the difference of the y-coordinates: 9 − 4 = 5 units.
x1_s2	Set 3: Which explanation is correct? Where is the point (0, 6)? [On the y-axis, because you move 0 units right and then 6 units up / On the x-axis, because the 0 comes first / At the origin, because one coordinate is 0 / On the x-axis, 6 units right of the origin]	**On the y-axis, because you move 0 units right and then 6 units up**	An x-coordinate of 0 means you do not move right at all, so the point sits on the y-axis, 6 units up.
x2_s2	Set 3: Find the mistake: Reyna says that for Rule A: start at 0, add 2, and Rule B: start at 0, add 6, each Rule B number is always 4 more than the Rule A number. What should she fix? [Each Rule B number is 3 times the Rule A number; “4 more” only works for the pair (2, 6) / Each Rule B number is 6 more than the Rule A number / Each Rule B number is 2 times the Rule A number / Nothing. 4 more is correct]	**Each Rule B number is 3 times the Rule A number; “4 more” only works for the pair (2, 6)**	Rule A: 0, 2, 4, 6. Rule B: 0, 6, 12, 18. (2, 6) is 4 more, but (4, 12) is 8 more. The rule that works for every pair is 3 times: 2 × 3 = 6, 4 × 3 = 12, 6 × 3 = 18.

FIX THE SENTENCE (2 POINTS EACH)

ID	Question	Answer	Why (explain to student)
s1_s0	Set 1: Fix this sentence: quinlan asked, where is the origin on the map	**Quinlan asked, "Where is the origin on the map?"**	Capitalize the name Quinlan, put quotation marks around his exact words, capitalize the first word of the quote, and put the question mark inside the closing quotation mark.
s2_s0	Set 1: Fix this sentence: The bison lifted it's head from the grass.	**The bison lifted its head from the grass.**	Its (no apostrophe) shows ownership: the head belongs to the bison. It's always means it is.
s3_s0	Set 1: Fix this sentence: The map shows geysers hot springs and waterfalls.	**The map shows geysers, hot springs, and waterfalls.** (also accepted: The map shows geysers, hot springs and waterfalls.)	Use commas to separate the items in a series: geysers, hot springs, and waterfalls.
s4_s0	Set 1: Fix this sentence: the points on the graph forms a straight line	**The points on the graph form a straight line.** (also accepted: The points on the graph formed a straight line.)	Capitalize the first word. The subject is points (plural), not graph, so use form, not forms. End with a period.
s1_s1	Set 2: Fix this sentence: ranger roderick said, always stay on the boardwalk	**Ranger Roderick said, "Always stay on the boardwalk."**	Capitalize Ranger Roderick (a title used with a name), put quotation marks around his exact words, capitalize the first word of the quote, and put the period inside the closing quotation mark.

ID	Question	Answer	Why (explain to student)
s2_s1	Set 2: Fix this sentence: The hikers packed there water bottles before sunrise.	**The hikers packed their water bottles before sunrise.**	Their shows ownership: the water bottles belong to the hikers. There tells where something is.
s3_s1	Set 2: Fix this sentence: After lunch the cousins plotted the footbridge on the grid.	**After lunch, the cousins plotted the footbridge on the grid.**	Put a comma after the opening phrase After lunch.
s4_s1	Set 2: Fix this sentence: ramona plot each point with a sharp pencil	**Ramona plots each point with a sharp pencil.** (also accepted: Ramona plotted each point with a sharp pencil.)	Capitalize the name Ramona. One person needs the verb plots, not plot. End with a period.
s1_s2	Set 3: Fix this sentence: ramona said, the x-coordinate always comes first	**Ramona said, "The x-coordinate always comes first."**	Capitalize Ramona, put quotation marks around her exact words, capitalize the first word of the quote, and put the period inside the closing quotation mark.
s2_s2	Set 3: Fix this sentence: Its a long walk to the overlook.	**It's a long walk to the overlook.**	It's means it is: it is a long walk. Use the apostrophe.
s3_s2	Set 3: Fix this sentence: Yellowstone stretches into Wyoming Montana and Idaho.	**Yellowstone stretches into Wyoming, Montana, and Idaho.** (also accepted: Yellowstone stretches into Wyoming, Montana and Idaho.)	Use commas to separate the items in a series: Wyoming, Montana, and Idaho.
s4_s2	Set 3: Fix this sentence: the rangers at the station checks the trail map every day	**The rangers at the station check the trail map every day.** (also accepted: The rangers at the station checked the trail map every day.)	Capitalize the first word. The subject is rangers (plural), not station, so use check, not checks. End with a period.

MATCH-UP ROUND (2 POINTS PER FIRST-TRY MATCH)

ID	Question	Answer	Why (explain to student)
match_s0_1	Set 1: Match 6 right, 2 up	**(6, 2)**	2 points if matched on the first try.
match_s0_2	Set 1: Match 2 right, 6 up	**(2, 6)**	2 points if matched on the first try.
match_s0_3	Set 1: Match The origin	**(0, 0)**	2 points if matched on the first try.
match_s0_4	Set 1: Match On the y-axis, 7 up	**(0, 7)**	2 points if matched on the first try.
match_s0_5	Set 1: Match Start at 0: add 3 and add 9	**Rule B = 3 × Rule A**	2 points if matched on the first try.
match_s1_1	Set 2: Match 4 right, 1 up	**(4, 1)**	2 points if matched on the first try.
match_s1_2	Set 2: Match 1 right, 4 up	**(1, 4)**	2 points if matched on the first try.
match_s1_3	Set 2: Match On the x-axis, 8 right	**(8, 0)**	2 points if matched on the first try.
match_s1_4	Set 2: Match Next after (2, 6), (4, 12)	**(6, 18)**	2 points if matched on the first try.
match_s1_5	Set 2: Match Start at 0: add 5 and add 10	**Rule B = 2 × Rule A**	2 points if matched on the first try.
match_s2_1	Set 3: Match 7 right, 3 up	**(7, 3)**	2 points if matched on the first try.
match_s2_2	Set 3: Match 3 right, 7 up	**(3, 7)**	2 points if matched on the first try.

ID	Question	Answer	Why (explain to student)
match_s2_3	Set 3: Match On the y-axis, 5 up	**(0, 5)**	2 points if matched on the first try.
match_s2_4	Set 3: Match Next after (3, 12), (6, 24)	**(9, 36)**	2 points if matched on the first try.
match_s2_5	Set 3: Match Start at 0: add 4 and add 2	**Rule B = half of Rule A**	2 points if matched on the first try.

BOSS BATTLE: THE JUNIOR RANGER MAP CHALLENGE (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
b1_s0	Set 1: *Quinlan and Ramona Plan a Loop Hike On Ranger Roderick's practice map, the Trailhead is at (2, 1), the Footbridge is at (2, 7), the Overlook is at (9, 7), and the Picnic Area is at (9, 1). The cousins walk along the grid lines from the Trailhead to the Footbridge, the Overlook, the Picnic Area, and back to the Trailhead. Each grid unit stands for 1/2 mile.* Two steps: how many grid units long is the whole loop? [26 units / 13 units / 42 units / 22 units]	**26 units**	Trailhead to Footbridge: 7 − 1 = 6 units. Footbridge to Overlook: 9 − 2 = 7 units. The loop is a rectangle, so the other two sides are also 6 and 7. 6 + 7 + 6 + 7 = 26 units. (13 counts only half the loop, and 42 multiplies 6 × 7.)
b2_s0	Set 1: Two steps: how many miles long is the whole loop? Miles:	**13** (also accepted: 13 miles, 13 mile, 13 mi)	The loop is 26 grid units. Each unit stands for 1/2 mile: 26 × 1/2 = 13 miles.
b1_s1	Set 2: *Quinlan and Ramona Race the Patterns Two hiking groups leave the same trailhead at the same time. Quinlan's group walks 2 miles every hour, and Ramona's faster group walks 3 miles every hour. Ranger Roderick writes the ordered pairs (miles for Quinlan's group, miles for Ramona's group): (0, 0), (2, 3), (4, 6), (6, 9).* Which ordered pair comes right after (6, 9)? [(8, 12) / (7, 10) / (9, 6) / (8, 11)]	**(8, 12)**	Quinlan's group adds 2 miles each hour (6 + 2 = 8) and Ramona's group adds 3 miles each hour (9 + 3 = 12). The next pair is (8, 12).
b2_s1	Set 2: Two steps: when Quinlan's group has walked 14 miles, how many miles has Ramona's group walked? Miles:	**21** (also accepted: 21 miles, 21 mile, 21 mi)	Quinlan's group reaches 14 miles after 14 ÷ 2 = 7 hours. In 7 hours, Ramona's group walks 7 × 3 = 21 miles. The point is (14, 21).
b1_s2	Set 3: *Ranger Roderick Fills the Water Tank Ranger Roderick fills an empty water tank at the ranger station at a steady rate. He records (minutes, gallons in the tank): (0, 0), (2, 15), (4, 30), (6, 45).* If the pattern continues, which ordered pair comes next? [(8, 60) / (8, 55) / (60, 8) / (7, 60)]	**(8, 60)**	The minutes go up by 2 and the gallons go up by 15 each time. 6 + 2 = 8 and 45 + 15 = 60, so the next pair is (8, 60).
b2_s2	Set 3: Two steps: the tank holds 120 gallons when it is full. How many minutes does it take to fill the empty tank? Minutes:	**16** (also accepted: 16 minutes, 16 minute, 16 min)	The tank gains 15 gallons every 2 minutes. 120 ÷ 15 = 8 groups of 2 minutes, and 8 × 2 = 16 minutes. The point is (16, 120).

Pre & Post sheet answer guide

Many items are open-ended. Grade for genuine understanding and complete sentences, not exact wording. Sheet: grade4-5/prepost/KCE_Math_PrePost_G4-5W21.pdf.

PRE-LESSON WARM-UP

#	Prompt	Model answer / grading note
1	Review: 7 × 5 × 2 bin; 240 cu in box, 10 × 6 base	a) 70 cubic feet b) 240 ÷ (10 × 6) = 240 ÷ 60 = 4 inches
2	Count by 4s and by 8s from 0 (first five numbers)	By 4s: 0, 4, 8, 12, 16. By 8s: 0, 8, 16, 24, 32.
3	Compare numbers in the same position	Each 8s number is 2 times the matching 4s number (16 = 2 × 8). Accept any reasonable pattern before the lesson.
4	What (3, 5) tells you to do	Move 3 right, then 5 up
5	Two number lines from 0 to 6; dot 2 right and 4 up	Axes meet at 0; the dot is at (2, 4). Accept any reasonable attempt.

POST-LESSON CHALLENGE

#	Prompt	Model answer / grading note
1	a) 7 right, 2 up b) Is (0, 6) on the x-axis or y-axis? c) (5, 4), move 3 left and 4 up	a) (7, 2) b) the y-axis (the x-coordinate is 0) c) 5 − 3 = 2 and 4 + 4 = 8, so (2, 8).
2	Ranger Station (2, 3) to Overlook (2, 9); 1/4-mile scale; there and back	9 − 3 = 6 units; 6 × 1/4 = 1 1/2 miles apart. There and back: 3 miles.
3	Rule A: start at 0, add 6. Rule B: start at 0, add 3	Pairs: (0, 0), (6, 3), (12, 6), (18, 9), (24, 12). Each Rule B number is half of the Rule A number.
4	(1, 3), (2, 6), (3, 9): which point shows 7 hours?	(7, 21), because y = 3 × x.
5	Ruben says (4, 1) and (1, 4) are the same point	Model: Ruben is wrong, because order matters. (4, 1) means 4 right and 1 up, but (1, 4) means 1 right and 4 up, so they are different points.

★ CHALLENGE

#	Prompt	Model answer / grading note
1	Rule A: start at 1, add 2. Rule B: start at 4, add 2	(1, 4), (3, 6), (5, 8), (7, 10). 3 more: both rules add 2, so the gap stays 3. “3 times” fails (1 × 3 = 3, not 4).